OPERATIONAL MANUAL & ARCHITECTURE REFERENCE

Flow Music MCP — Claude + Gemini + Google Flow Music

End-to-End AI Music Video Storyboard & Video Production Pipeline (flow-music-mcp)

1. SYSTEM ARCHITECTURE OVERVIEW

1. Claude Code / Desktop

Acts as the AI Director & Host. Connects to `flow-music-mcp` over JSON-RPC 2.0 stdio to orchestrate audio analysis, scene planning, edits, and video generation in plain English.

2. Google Gemini API

Audio + Image Engine: Extracts exact lyrics & `MM:SS` timestamps from the MP3, generates Character Model Sheets, and renders consistent starting scene keyframes.

3. Google Flow Music

Video Engine (flowmusic.app): Driven via Chrome DevTools Protocol (CDP). Opens a new session, attaches the MP3 & scene frames (or chains last frames), and generates all scene MP4s.

2. UNDERSTANDING THE TWO FOLDERS (ENGINE VS. SONG WORKSPACE)

Keeping the MCP server separate from your song folders lets you create music videos for unlimited songs cleanly:

Folder A: The MCP Server Folder (Set Up Once)

Example Path: `C:\path\to\flow-music-mcp`

This folder holds the TypeScript/Node.js MCP engine. You only run `npm install`, `npm run build`, and `npm run login` here once. **Do not place your MP3s here.**

Folder B: Your Song / MP3 Folder (One Per Song)

Example Path: `C:\Users\You\Music\MySong`

Place the `.mp3` of the song you are making a music video for inside any folder. When you tell Claude to connect to this folder, all outputs are saved right alongside your MP3:

```
<your-song-mp3-folder>/
├── song.mp3                # Your input MP3 audio track
├── STORYBOARD.md           # Human-editable Markdown table (Lyrics, Bible, Scenes — two-way synced!)
├── storyboard.json         # Machine-readable storyboard state
├── characters/             # Gemini-generated character model sheets (e.g., leo_indoor.png, leo_rain.png)
├── scenes/                 # Gemini-generated starting keyframe images (shot-01.png, shot-02.png, ...)
└── videos/                 # Final video clips downloaded from Google Flow Music (shot-01.mp4, ...)
```

3. PART I — ONE-TIME TERMINAL SETUP (BEFORE OPENING CLAUDE)

STEP 1 **Install & Build the MCP Server:**

```
git clone https://github.com/purnanandsamant/flow-music-mcp.git
cd flow-music-mcp
npm install
npm run build
```

STEP 2 **Register with Claude Code (or Claude Desktop):**

```
claude mcp add --scope user flow-music -- node "C:\Users\Maddy\Documents\Python\Data_Projects\flow-music-mcp\dist\index.js"
```

(Using `--scope user` makes the MCP available across every folder on your computer.)

STEP 3 **Sign in to Google Flow Music (Once):**

```
npm run login
```

Opens a dedicated Chrome profile at **https://www.flowmusic.app/**. Click **Login** → **Continue with Google**, check the box for **Google One AI benefits**, and leave this Chrome window open.

4. PART II — STEP-BY-STEP WORKFLOW INSIDE CLAUDE CODE

Launch Claude Code in your terminal by running `claude` (verify connection anytime with `/mcp`). Everything from here is conversational:

PHASE 0 **Save Your Gemini API Key (First Time Only)**
If `GEMINI_API_KEY` is not set in your system environment variables, tell Claude:

> *"Set my Gemini API key to AIza..."*

Claude calls `gemini_set_api_key` and stores it locally in `~/.flow-music-mcp/gemini-key`.

PHASE 1 **Connect to Your MP3 Folder & Identify Lyrics + Timestamps**
Point Claude to the folder containing your song's `.mp3`:

> *"Connect to my song folder at C:\path\to\my-song-folder, identify the lyrics and timestamps from the MP3 using Gemini, and draft sub-10-second storyboard scenes."*

Claude calls `project_connect` and `music_identify_lyrics`. Gemini listens to the MP3, extracts every vocal line and instrumental section with `MM:SS-MM:SS` timestamps, and drafts sub-10s scenes in `STORYBOARD.md`.

PHASE 2 **Scene Frame Continuity Modes (Start Frame vs. Previous Last Frame)**
Each scene in the storyboard uses one of two continuity modes:

- `generate_start_frame` — Gemini generates a brand-new starting keyframe image (`scenes/shot-NN.png`) for new camera angles, new scenes, or new environments.
- `use_previous_last_frame` — Instructs the pipeline to extract the **exact final frame** from the previous scene's video so the new clip continues in one unbroken motion.

PHASE 3 **Review & Edit Scenes (Via Chat or Direct File Editing)**
You can refine any scene, timestamp, camera angle, or `frame_mode` in two ways:

> *"Change Shot 4 to use_previous_last_frame from Shot 3, and update Shot 5 to a low-angle tracking shot."*

Or open `STORYBOARD.md` in your text editor, edit the Markdown table directly, save, and tell Claude: "*Reload my storyboard.*"

PHASE 4 **Generate Character Sheets & Scene Keyframe Images with Gemini**
First, lock your character model sheets (with hex color codes and outfit variations), then generate all scene images:

> *"Define our lead character and generate their character reference sheet in Gemini. Once done, generate all starting scene images."*

Claude calls `storyboard_upsert_character` → `gemini_generate_character_image` (saving to `characters/`), and then `gemini_generate_scene_images` (automatically passing the character sheet into Gemini so the character looks identical in every scene image inside `scenes/`). Review the PNGs in `scenes/` and ask Claude to regenerate any shot you want to tweak!

PHASE 5 **Start a New Google Flow Music Session & Generate All Videos**
Once all scene images in `scenes/` are approved, tell Claude:

> *"Check my Google Flow Music status, start a new music video session with my MP3, initiate the storyboard, and generate all scene videos."*

Claude calls `flow_music_status` → `flow_music_new_session` (opens **Music videos** → **New music video** in `flowmusic.app` and uploads your MP3 via **Add** → **Audio**) → `flow_music_generate_videos` (uploads each scene's starting image or extracted previous final frame, submits the scene video prompt, waits for completion, and downloads `videos/shot-01.mp4 ... shot-NN.mp4`).

Non-Blocking Task Tickets (No 60s Timeout): Video rendering in Google Flow Music takes 1–3 minutes per shot. Any tool call running longer than 45 seconds automatically returns a `task_id` while continuing in the background, and Claude uses `flow_music_wait` to collect the finished videos without ever timing out.

5. COMPLETE MCP TOOLS, RESOURCES & PROMPTS REFERENCE

Capability Name	Phase / Type	Description & Behavior
`project_connect`	Phase 0 (Tool)	Connects to the folder containing your `.mp3`, probes audio duration, and initializes or loads `storyboard.json`, `STORYBOARD.md`, `characters/`, `scenes/`, and `videos/`.
`gemini_set_api_key`	Phase 0 (Tool)	Saves your Google AI Studio `GEMINI_API_KEY` to `~/.flow-music-mcp/gemini-key`.
`music_identify_lyrics`	Phase 1 (Tool)	Uploads the MP3 to Gemini multimodal audio to extract timestamped lyrics (`MM:SS-MM:SS`), song sections, BPM/mood, and drafts initial sub-10s scenes.
`storyboard_get`	Phase 2–3 (Tool)	Returns the full storyboard state and automatically syncs any manual edits made directly in `STORYBOARD.md`.
`storyboard_set_production_bible`	Phase 2–3 (Tool)	Sets the Mandatory Style Anchor, aspect ratio, and Environment/Lighting settings (Kelvin lighting, hex palettes, camera lens style).
`storyboard_upsert_character`	Phase 2–4 (Tool)	Creates or updates a Character Model Sheet spec (proportions, hair/eyes, mandatory Character Anchor string, signature props, and outfits).
`storyboard_set_scenes`	Phase 2–3 (Tool)	Sets the complete list of sub-10-second scenes with `frame_mode` (`generate_start_frame` or `use_previous_last_frame`).
`storyboard_edit_scene`	Phase 3 (Tool)	Updates, inserts, or deletes an individual shot (timestamps, lyrics, camera/action, `frame_mode`, prompts, character/outfit refs).
`calculate_scene_frames`	Phase 2–3 (Tool)	Calculates exact animation frame count for a shot given duration in seconds and FPS.
`gemini_generate_character_image`	Phase 4a (Tool)	Generates character reference sheet / outfit PNGs via Gemini Image Generation into `<folder>/characters/`. Locks appearance across outfits.
`gemini_generate_scene_images`	Phase 4b (Tool)	Generates starting scene keyframes (`<folder>/scenes/shot-NN.png`) via Gemini Image Generation, attaching locked character reference sheets.
`flow_music_login`	Phase 5 (Tool)	Opens the dedicated Chrome profile at `https://www.flowmusic.app/` for one-time user sign-in.
`flow_music_status`	Phase 5 (Tool)	Checks Google Flow Music login state, active URL, credit/plan text, visible controls, and running background tasks.
`flow_music_new_session`	Phase 5a (Tool)	Opens **Music videos** → **New music video** (or a new Producer Session) in Google Flow Music, uploads the MP3 (**Add** → **Audio**), and selects the video model.
`flow_music_generate_videos`	Phase 5b (Tool)	Initiates the storyboard in Google Flow Music, attaches each shot's Gemini start image (or extracted previous last frame), generates videos, and downloads MP4s to `<folder>/videos/`.
`flow_music_wait`	Phase 5 (Tool)	Waits on background task tickets for operations running longer than 45 seconds.
`flow_music_inspect_and_act`	Phase 5 (Tool)	Inspects live DOM/ARIA elements, takes screenshots, or performs custom clicks/fills/uploads in the Google Flow Music Chrome window.
`config://studio/character-lock`	MCP Resource	Read-only resource returning the active project's Mandatory Style & Character Consistency Lock.
`storyboard://current/markdown`	MCP Resource	Read-only resource returning the live `STORYBOARD.md` Production Bible.
`music_video_production_workflow`	MCP Prompt	Reusable slash prompt guiding Claude through all 5 phases for a given MP3 folder.
`review_shot_prompt`	MCP Prompt	Verifies any shot prompt against Character & Style Consistency rules.